



# On the usage of pulsator data within the load spectra assessment of gears

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## ABSTRACT

Tooth root fatigue fracture is one of the most critical gear failure modes. This is due to the fact that it may cause a sudden interruption of the power flow within the gearbox, which typically results in the complete failure of the transmission. Therefore, a crucial design aspect is the accurate characterization of the gear bending fatigue strength. Here, designers can rely on experimental tests in order to obtain data related to their specific industrial reality.

Within this context, pulsator test, also called Single Tooth Bending Fatigue (STBF) test, is the most adopted experimental methodology. As such tests are performed only on selected teeth within a testing device and not on all teeth of a running gear pair, the pulsator experimental outcome has to be elaborated before using it in the actual design phase.

Depending on the elaboration methods, several SN curves can be obtained, and those curves will differ from the one predicted by the standards (e.g. ISO 6336 and ANSI AGMA 2001-D04). Furthermore, different SN curves will provide a different assessment when load spectra are considered. Here, the usage of data obtained from pulsator tests is analyzed within the phase in which load spectra are considered while designing a gear. In particular, this article discusses the differences between various elaboration methodologies and an assessment based on standardized curves.

## 1. Introduction

The design process of a gear train involves several steps. It begins with an initial pre-sizing phase, where key parameters such as module and face width are determined. Subsequently, more detailed stages are performed in order to obtain the most suitable geometry, hence defining the actual profile (i.e. the tools parameters) together with profile modifications (e.g. crowning, tip and root relief). Within this refinement phase, several phenomena are considered: components deformability, lubrication, noise. Furthermore, the fact that the gearbox typically does not operate at a fixed load, but under a certain load spectrum, must be taken into account.

This last aspect is addressed by ISO 6336-6:2019 [1], whose methodology is also recommended by ANSI AGMA 2001-D04 [2]. On one hand, ISO 6336-6:2019 [1] proposes a calculation method for the purpose of determining an equivalent load to be used during the pre-sizing phases. On the other hand, it allows to estimate a safety coefficient with respect to the applied spectrum. This standard applies to both macropitting and tooth root fatigue failure. The basis of this standard lies in three elements: a known load spectrum, a defined damage accumulation law (i.e. Palmgren-Miner rule [3,4] for the standards [1,2]), and a known gear SN curve. This curve can be

derived within the normative framework (i.e. [2,5–7]). However, in order to move away from the generic nature of the standards, hence using data related to their specific industrial reality, designers can rely on experimental tests with a view to obtaining such SN curve (as also suggested by the standards themselves [2,5–7]).

In the case of tooth root fatigue failure, an experimental SN curve can be obtained through various methods, including testing notched specimens (e.g. [8–14]), pulsator tests (e.g. [15–33]), also known as Single Tooth Bending Fatigue (STBF), and tests on running gears (e.g. [29,34,35]). Among these, the most adopted method is the pulsator one because it employs a representative specimen (the teeth of a spur gear) and it does not require a specialized testing equipment, considering that a universal testing machine is sufficient [36]. However, it should be noted that pulsator tests are typically conducted on the teeth (and not on the complete gear) without gear meshing. Therefore, pulsator test results need to be further processed in order to use the experimental evidence for the assessment of an actual gear pair (e.g. [15,17,31,37–40]).

The final outcomes of these elaborations are SN curves, defined at a certain reliability level (typically 1% of failure probability). Those curves can then be used for spectrum verification of a gear pair.

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As first analysis, such curves may differ from the ones suggested by the standards (i.e. [6,7] for tooth root fatigue fracture) because of the inherent material variations. Furthermore, different curves can be obtained on the basis of the adopted elaboration method. Hence, the assessment outcome might differ as well.

This article addresses the above-mentioned topic: the usage of the experimental evidence obtained from pulsator tests, in addition to the subsequent elaboration, within the framework of load spectra assessments. Moreover, it discusses how the assessment results differ from the one obtained adopting the curves presented by the standards.

For the purpose of achieving this, first of all the test methodology and the data processing methods are presented, followed by a brief overview of how the reference ISO standard suggests performing spectrum verification. Consequently, two case studies based on different load spectra for different geared applications are presented (i.e. [1,41,42]), followed by a discussion of the results.

The analysis presented here is based on five sets of experimental tests (named A, B, C, D and E), which have been conducted by the authors using a symmetric pulsator on case-hardened gears. Such dataset is described in [43].

## 2. Pulsator tests as a test for tooth root fatigue fracture

All the tests investigating the gear strength present a major challenge: the specific failure mechanism of interest (e.g. tooth root fatigue fracture) must be highlighted with respect to all others (e.g. macro and micro pitting, scuffing, and wear). This problem is faced in two ways: either by designing a test specimen (i.e. a gear) to be used inside an experimental gearbox that results to be weaker for the failure mechanism of interest (e.g. [44,45]) or, as in the case of tooth root fatigue fracture, by adopting a testing methodology where only one failure mechanism is present.

Pulsator tests utilize the latter idea. Accordingly, tooth root bending fatigue is replicated by applying an alternating load at a fixed position on the flanks, typically adopting a uniaxial testing machine. Various test configurations can be used, including applying torque to a geared shaft against a fixed anvil (e.g. [15]), three-point bending (e.g. [46,47]), and single-tooth (asymmetric, e.g. [16,20,24,31]) or two-tooth (symmetric, e.g. [18,23,26–29]) pulsator configurations. These two last testing methodologies are the most widely adopted. For more details about pulsator test methods, the interested reader can refer to [31,43].

## 3. Elaboration of pulsator tests

The lack of gears meshing within pulsator tests implies differences between the test itself and the real case, where gears rotate under load. On the one hand, those differences are related to the distinct tooth root stress histories that involve a different accumulation of fatigue damage (e.g. [40,48–51]). On the other hand, there are dissimilarities of statistical nature. Indeed, in pulsator tests, the failed teeth are predetermined, whereas, during gear meshing, the failure of a gear is represented by the failure of its weakest tooth. Therefore, because of the above differences, it is necessary to process the test data before using them in the design phase.

It is not surprising that the issue of pulsator data elaboration arises with the test itself. In fact, as early as 1964, when Seabrook and Dudley presented their test results on gear teeth, they observed that experimental data needed to be processed in order to obtain representative data for the real case [15]. Subsequently, over the years, other data processing methodologies have been proposed, such as the one by Rao and McPershon [52,53] and the FVA 304 [17] by Stahl. This latter evaluation method for pulsator test data is well-established and suitable for determining strength values (i.e. the long life) in a reliable way [54,55]. In addition to these, there have also been some recent works by Bonaiti et al. (i.e. [38–40]). It is also worth mentioning a

more recent methodology presented by Hong [31], which still needs to be improved, according to Hong himself.

All those works aim to estimate a SN curve referred to the gear, typically for 1% failure probability, starting from the experimental results provided by pulsator tests. Hence, they all address three different aspects.

Firstly, as in all the cases related to fatigue tests, they elaborate the fatigue test results. Rao and McPershon [52,53] adopt a “modified staircase sequence” to estimate both limited life and long life regions. The FVA 304 [17] adopts a staircase sequence for the long life region and a line fitting for the limited life region, which are then joined in order to describe the whole SN curve with a bi-linear description. In [38,39], Maximum Likelihood (ML) is adopted to describe a unique SN curve adopting the Spindel–Haibach model [56] while Hong [31] also adopts ML, but to estimate the parameters of the Pascual and Meeker model with non-constant standard deviation and a fatigue limit (i.e. [57]).

Secondly, they translate the test data from the tested teeth to the whole gear. Typically, corrective coefficients based on typical gear scattering (as in [17]) or statistical considerations are adopted (as in [31,38,39,52,53]).

Lastly, the different loading conditions between the pulsator test and the actual meshing gear scenario need to be taken into account. This last aspect is addressed by corrective coefficients, for instance as the one defined by Rettig [37] (as in [37], where Rettig’s coefficient is applied on the long life range) or other approaches, such as the use of allowable stress range diagrams, as in [52,53].

All these methodologies suggest a similar scaling of the pulsator data, or rather, they provide a similar result, especially in the long life region of the gear SN curve [43].

## 4. Adopted models

Among all methodologies, this article focuses on the approach described in FVA report 304 [17] and the one described in [38,39]. Both have been developed for a symmetric test rig configuration. Those methodologies are described deeply in [43] as well as in their original sources (i.e. [17,38,39]). The curves deriving from the FVA report 304 [17] will be called “FVA, D.S” and “FVA, F.L.” depending on the presence of a double slope or of a fatigue limit in the long life region. On the other hand, the ones derived from [38,39] will be called “S. H., 2T” and “S. H., STBF” following the nomenclature and the statistical consideration proposed in [43].

However, both methods describe the SN curve with a bilinear (or trilinear) solution; consequently, the SN curve is expressed as a series of inclined and/or horizontal lines. This assumption represents a mathematical simplification of the fatigue problems [58], which results in a strict distinction of the different regions of the SN curve (i.e. the limited life and the long life region), showing no transition between them.

Hence, in order to overcome this statistical limitation, the Random Fatigue Limit model proposed by Pascual and Meeker [59] and the three-parameters Weibull model proposed by Castillo and Fernández-Canteli [58,60] have been included within the framework of method 2 [43]. ML is used as a means to estimate the model parameters (i.e. the experimental SN curve) of Pascual and Meeker and Profatigue (i.e. a dedicated software [60]) for the Castillo and Fernández-Canteli ones. Statistics of Extremes (SoE) is used to translate the results from the experimental evidence to the gear. Those models have been considered because of their strong statistical background, their relatively low number of parameters and their wide usage. The curves obtained from those two statistical models will be called “R.F.L.” and “3 P. WBL” respectively.

A short description of the models is included in [Appendix](#).

Finally, as suggested by the FVA report 304 [17], the effect of the different loading conditions is taken into account by applying Rettig’s

**Table 1**

Summary of the adopted models. The “Statistical difference” and “Fatigue difference” columns refer to how the discrepancies between the experimental data (i.e. fixed teeth loaded by anvils) and the real case (i.e. gears meshing under load) have been considered.

| Name        | Model  | SN shape                          | Parameter estimation   | Statistical difference | Fatigue difference                                     |
|-------------|--------|-----------------------------------|------------------------|------------------------|--------------------------------------------------------|
| FVA, F.L.   | [17]   | Bi-linear, with fatigue limit     | FVA 304                | FVA 304                | $f_{P \rightarrow MG}$ on $\sigma_{(F0\infty,50\%,P)}$ |
| FVA, D.S.   | [17]   | Tri-linear, with double slope     | FVA 304                | FVA 304                | $f_{P \rightarrow MG}$ on $\sigma_{(F0\infty,50\%,P)}$ |
| S. H., STBF | [56]   | Bi-linear                         | ML                     | SoE                    | $f_{P \rightarrow MG}$ on $\sigma_e$                   |
| S. H., 2T.  | [56]   | Bi-linear                         | ML                     | SoE                    | $f_{P \rightarrow MG}$ on $\sigma_e$                   |
| R.F.L.      | [59]   | Continuous, with fatigue limit    | ML                     | SoE                    | $f_{P \rightarrow MG}$ on $\gamma$                     |
| 3 P. WBL    | [58]   | Continuous, with horizontal limit | ProFatigue             | SoE                    | $f_{P \rightarrow MG}$ on $C$                          |
| ISO, F.L.   | [6,17] | Bi-linear, with fatigue limit     | ISO 6336-6 and FVA 304 | FVA 304                | Acc. to FVA 304                                        |
| ISO, D.S.   | [6,17] | Bi-linear                         | ISO 6336-6 and FVA 304 | FVA 304                | Acc. to FVA 304                                        |

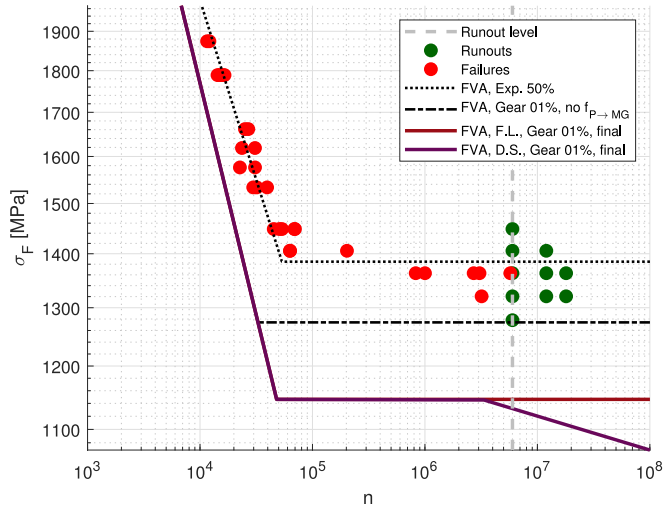


Fig. 1. Evaluation of SN curves for exemplary campaign B adopting the FVA report 304 [17].

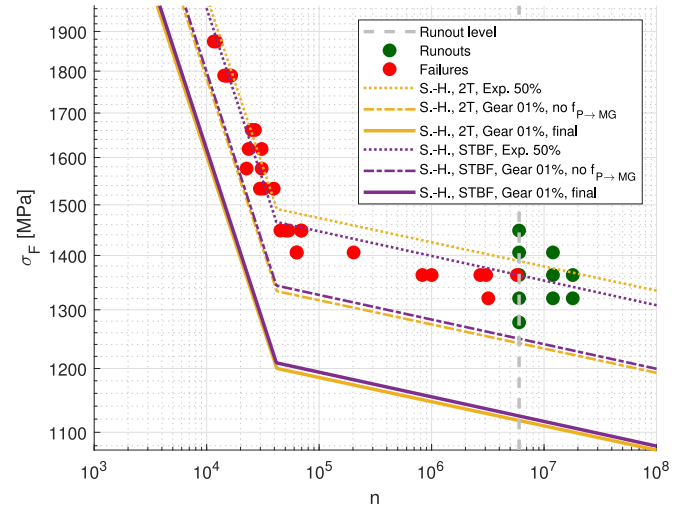


Fig. 2. Evaluation of SN curves for exemplary campaign B adopting the Spindel-Haibach model [56].

coefficient [37] (i.e.  $f_{P \rightarrow MG}$ ) to the parameter (of the different models) that resembles the fatigue limit. This coefficient, equal to 0.9, has been the topic of recent research [48–51], finding that it represents a conservative value for considering the effect of the different accumulation of fatigue damage. It must be highlighted that Rettig’s coefficient [37] has been mainly validated only for the long life regime while in the limited life range further effects should be expected.

In order to take into account the difference between the pulsator test results and the ISO 6336 framework, two additional curves need to be considered. Those curves are defined by adopting  $\sigma_{(F0\infty,1\%,MG)}$  (i.e. the gear fatigue limit in meshing gear condition according to the FVA report 304 [17]) as the base for estimating  $\sigma_{Flim}$  taking into account the values assumed by  $Y_{NT}$ ,  $Y_{\delta relT}$  and  $Y_{RrelT}$  (i.e. ISO 6336-3 [6] corrective coefficients) in the long life region. From such  $\sigma_{Flim}$ , considering the two extreme values of  $Y_{NT}$  in the long life region, together with the different values assumed by  $Y_{\delta relT}$  and  $Y_{RrelT}$  in the two regions of the SN curve, two different SN curves are derived. Those last two curves will be called “ISO, D.S” and “ISO, F.L.” depending on the presence of a double slope or of a fatigue limit in the long life region.

Table 1 summarizes all the adopted models, while Figs. 1–4 show their application to an exemplary campaign. Fig. 5 depicts all the final curves estimated for such campaign. All the curves are defined at 1% probability of gear failure.

## 5. ISO-based load spectra assessment

ISO 6336-6 [1] allows to take into account the presence of load spectra within the design/assessment phases of a gear train. To accomplish this, the standard requires three different tools.

Firstly, it is necessary to determine the load spectrum and the standard allows for the use of both measured spectra and estimated

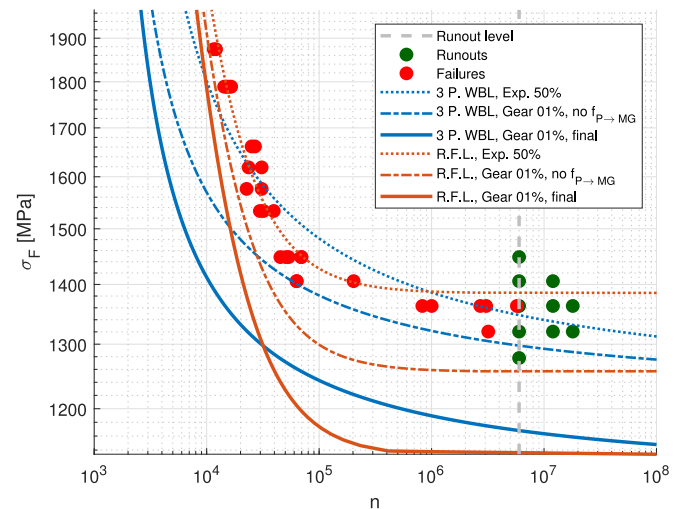


Fig. 3. Evaluation of SN curves for exemplary campaign B adopting the Pascual and Meeker [59] and the Castillo and Fernández-Canteli [58] models.

spectra. This spectrum should then be divided into approximately 64 bins.

Secondly, this spectrum should be compared with the SN curve related to the specific gear pair. In the ISO context, the curve can be obtained experimentally or, alternatively, by utilizing typical endurance strength values specified in ISO 6336-5 [7], along with specific corrective factors provided in ISO 6336-2 [5] and ISO 6336-3 [6].

The last tool is a damage law, and among the various damage laws (e.g. [61–63]), the Palmgren-Miner linear damage accumulation

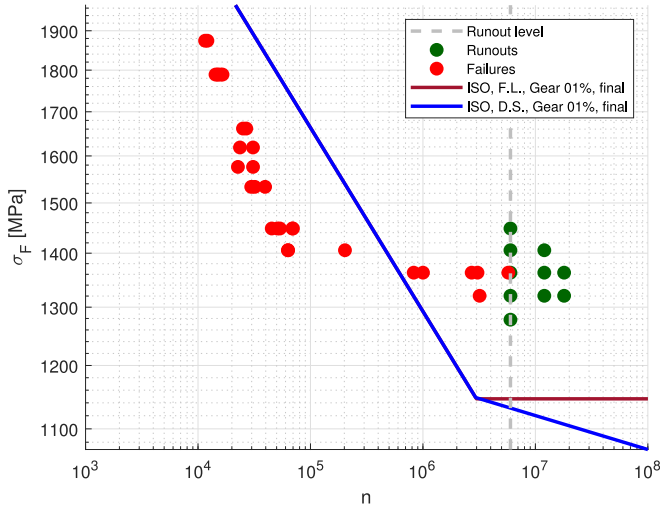


Fig. 4. Evaluation of ISO 6336-3 [6] SN curves for exemplary campaign B.

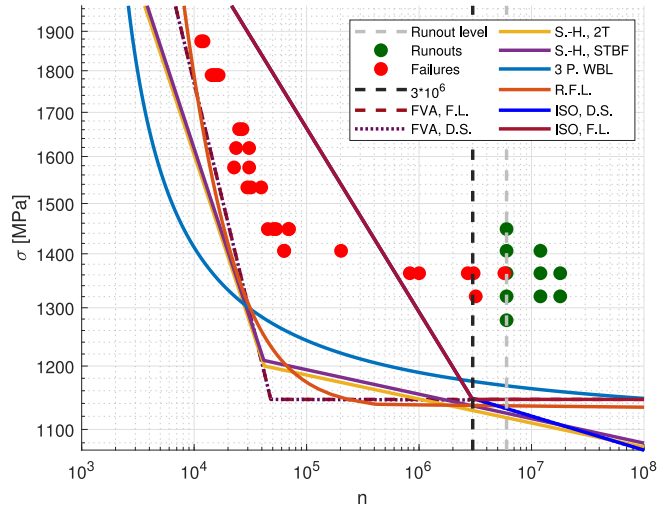


Fig. 5. Final curves estimated for exemplary campaign B.

rule [3,4] is chosen due to its simplicity. Accordingly, the cumulated damage  $U$  can be calculated as:

$$U = \sum_i \frac{n_i}{N_i} = \sum_i \frac{n_i}{N_{(\sigma_i)}} \quad (1)$$

where  $n_i$  is the number of cycles associated to the  $i$ th bin and  $N_{(\sigma_i)}$  is the maximum number of cycles that can be withstood at the  $\sigma_i$  stress level. If the total cumulated damage  $U$  is greater than one, the assessment is not satisfied; that is, the failure of the component can be expected [1].

ISO 6336-6 [1] proposes a method for the calculation of a safety factor  $S$  by scaling the spectra on the  $\sigma$  direction of the SN curve. Accordingly,  $S$  is defined as the scaling factor that leads to  $U = 1$ . Fig. 6 shows a graphical example of this calculation method based on Case-1 load spectrum: the original spectrum is scaled by  $S$ , defining the load spectrum that leads to  $U = 1$ .

## 6. Case 1: Industrial hoist spectrum

Within ISO 6336-6 [1] it is possible to find also an example of a load spectrum. It is based on a gear pair belonging to a reduction stage of a gearbox for an industrial hoist. This load spectrum has been measured over 70 days of loading and then extrapolated in order to be representative of 30 years of operation. The standard reports gear

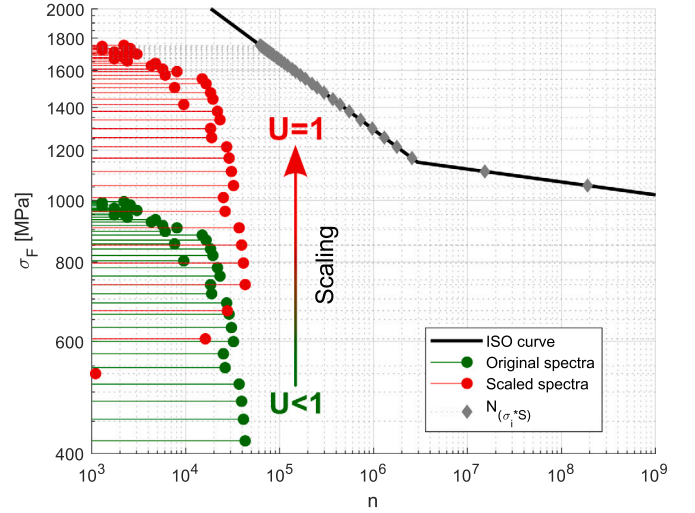


Fig. 6. Example of load spectra assessment according to ISO 6336-6 [1].

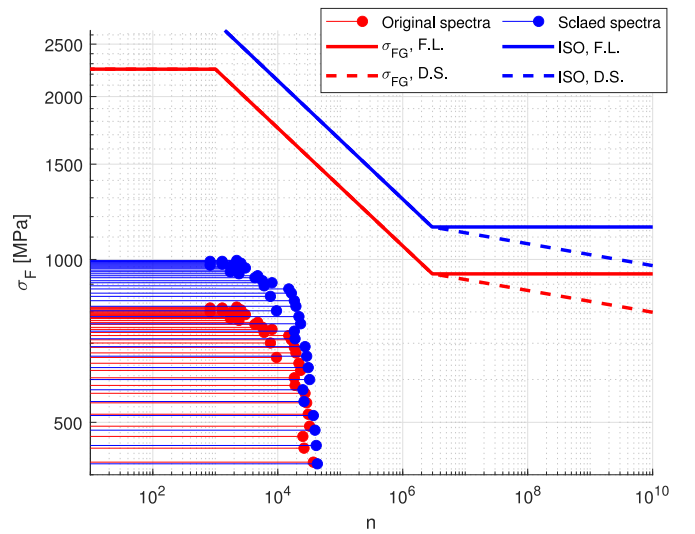


Fig. 7. Example of scaling procedure applied to the exemplary campaign B.

geometry, material and all the information from which two SN curves have been derived considering the two extreme values of  $Y_{NT}$  in the long life as well as the load spectrum (i.e.  $\sigma_{FG}$ , D.S. and  $\sigma_{FG}$ , F.L. in the graphs).

The five test campaigns here considered have been carried out on case-hardened gears subjected to different treatments and made of unidentical materials. Therefore, they will differ in strength values from one another as well as from the one of the gear presented in ISO 6336-6 [1]. To overcome this problem, for the different tests, the spectrum has been scaled. The scaling was made according to the ratio between the fatigue limit calculated according to FVA 304 [17] (i.e.  $\sigma_{(F0\infty,1\%,MG)}$ ), which is the basis for the definition of the “ISO, D.S.” and “ISO, F.L.” curves, and the fatigue limit of the gear defined in ISO 6336-6 [1] and calculate following ISO 6336-3 [6] (i.e.  $\sigma_{Flim} * Y_{ST} * Y_{\sigma relT} * Y_{RelT}$ ). Fig. 7 schematizes the application of the above scaling procedure. Since  $\sigma_{(F0\infty,1\%,MG)}$  is higher than the fatigue limit of the gear present in the original document [1], the spectrum is scaled up, while leaving the relative positioning between the load levels and the SN curve unaltered. This procedure is applied to all five campaigns. In this way, the differences between the different data sets are removed, hence highlighting only the differences between the applied statistical models.

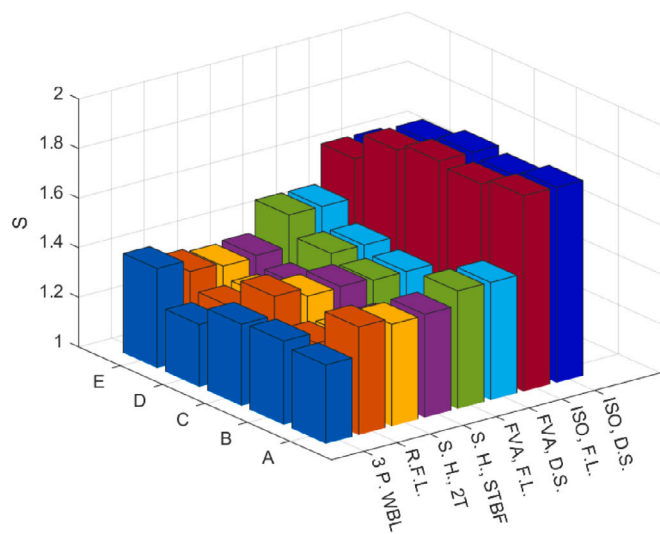


Fig. 8. Calculated safety factors for Case 1.

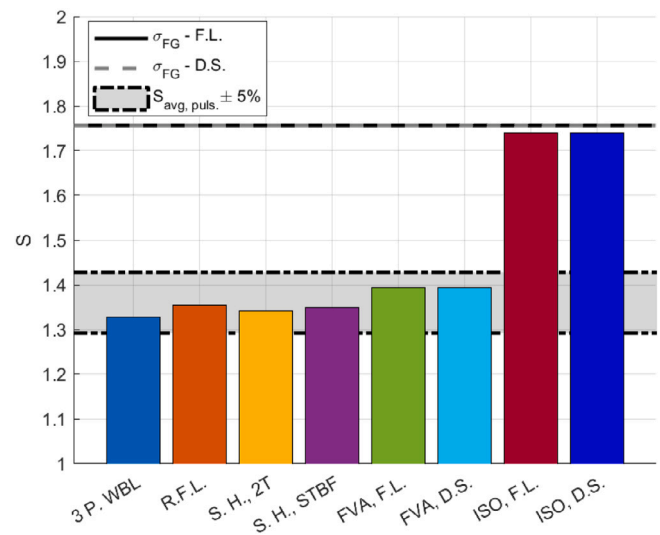


Fig. 9. Average safety factors for Case 1.

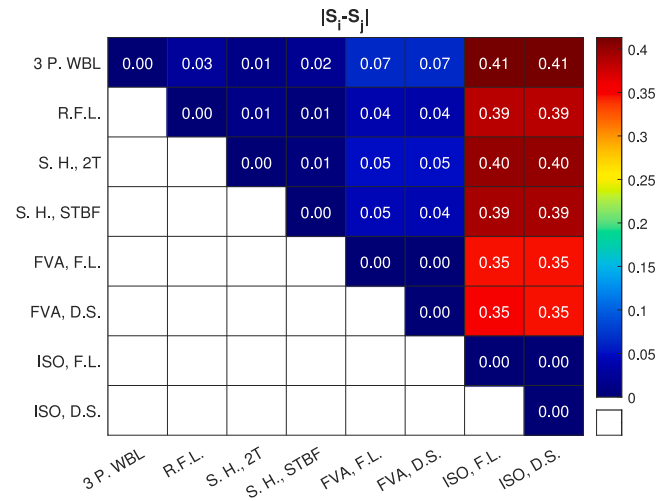


Fig. 10. Comparison between the different average safety factors.

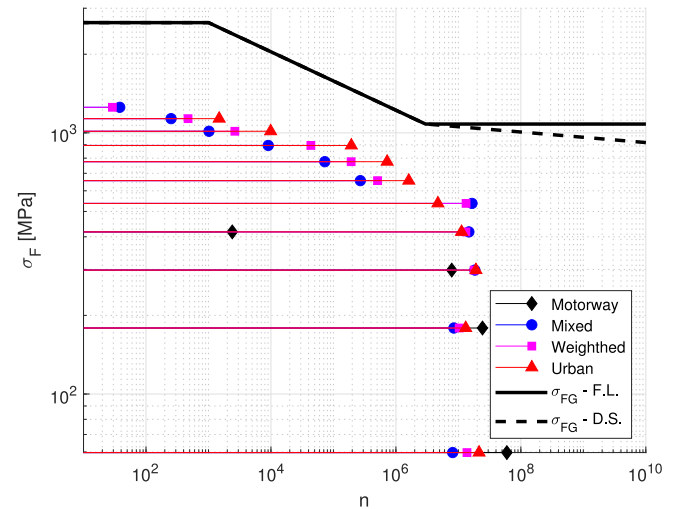


Fig. 11. Curà's load spectra [41,42].

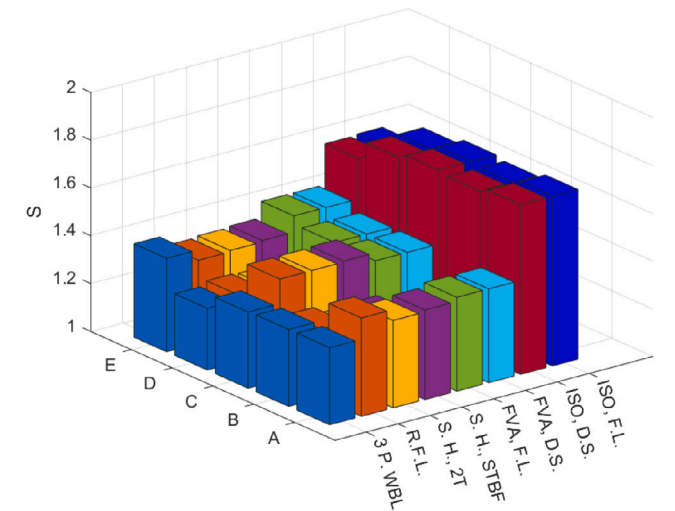


Fig. 12. Calculated safety factors for Case 2, "Weighted" spectra.

Fig. 8 shows the calculated safety factors for all materials and models, Fig. 9 reports the safety factors, averaged with respect to the campaigns, and Fig. 10 illustrates the difference between the various average safety factors.

## 7. Case 2: truck spectra

The second case study here examined is related to load spectra measured experimentally on the input gear of a transfer box for industrial truck by Curà [41,42]. In her works, Curà presented 4 different spectra, namely "Motorway", "Mixed Route and Mountain" (hereinafter called "Mixed"), "Urban route" (hereinafter called "Urban") and "Total weighted mission" (hereinafter called "Weighted"). Three of these (i.e. "Motorway", "Mixed" and "Urban") have been measured along 100-km road trips while "Weighted" is a weighted average of all three. Curà's studies also report geometry and material in a similar way to Case 1. Hence, both spectra (in terms of stress) and two SN curves (related to the gears under consideration) can be reconstructed. The negative parts of the spectra have not been considered. Fig. 11 reports the SN curves as well as the load spectra.

Similarly to the previous case, the safety factor of each spectrum, material and model can be calculated. Again, the spectra have been

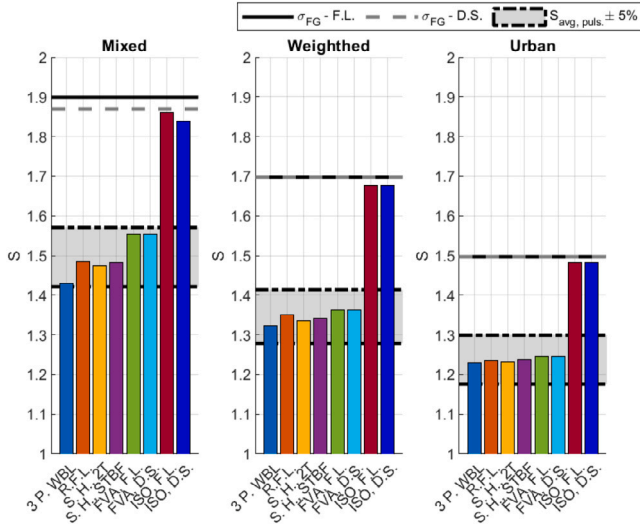


Fig. 13. Average safety factors for Case 2.

scaled as a means to keep the relative positioning between spectra and curves unaltered. Fig. 12 reports the calculated safety factors for the “Weighted” case, while Fig. 13 shows the results (averaged over the different materials). The spectrum “Motorway” has not been reported because it shows a very high safety factor ( $>3$  for all cases). Indeed, as reported in Fig. 11, its highest stress bin is substantially lower than the fatigue limit.

Regardless, a safety factor has a purely “engineering” meaning. In this specific case, we can supply the calculation with a physical sense by not working on  $S$ , but by estimating, still remaining within the Palmgren-Miner model [3,4], the expected lifetime of the truck or, in other words, the lifetime of this particular component in terms of kilometers. This could be achieved by calculating the damage  $U$  related to every single repetition of the spectrum, out of which the number of repetitions  $N_r$  is derived in the following way:

$$N_r U = 1 \quad (2)$$

$N_r$  represents how many times a particular mission can be repeated before the failure of 1 gear pair over 100 (as the curves are defined at 1% probability of failures). Knowing that each spectrum has been measured under actual on-road conditions over 100 km, it is possible to estimate the number of kilometers. Here, results are shown in terms of thousands of kilometers in order to reduce the number of digits.

Fig. 14 shows the estimated lifetimes for all materials and models considering the “Weighted” spectrum. On the other hand, Fig. 15 reports the logarithmic mean of the lifetimes for all the spectra and Fig. 16 illustrates the difference between the variously estimated lifetimes for the “Weighted” spectra. Again, the spectrum “Motorway” has not been reported.

## 8. Results

The five experimental pulsator test campaigns performed on case-hardened gears reported in [43] (named A, B, C, D and E) have been used to derive several SN curves referred to the gears, under meshing conditions and at a reliability level of 99% (i.e. at a 1% probability of failure). All those curves constituted the basis for spectrum verification according to ISO 6336-6 [1] and are summarized in Table 1. The SN curves derived according to ISO 6336-6 [1] for the gears reported in the original sources of the spectra (i.e. [1,41,42]) have been considered as well. The adopted spectra relate to industrial and automotive applications. Specifically, the first belongs to an industrial hoist [1] while the latter to the transfer box of an industrial truck [41,42].

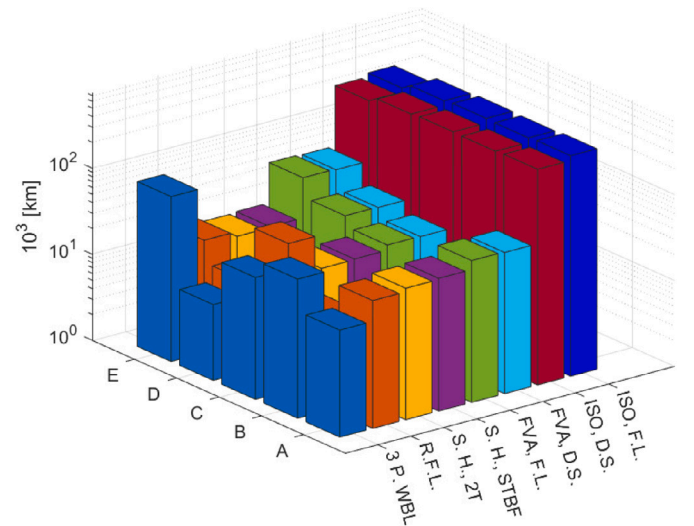


Fig. 14. Estimated lifetimes for the “Weighted” spectrum.

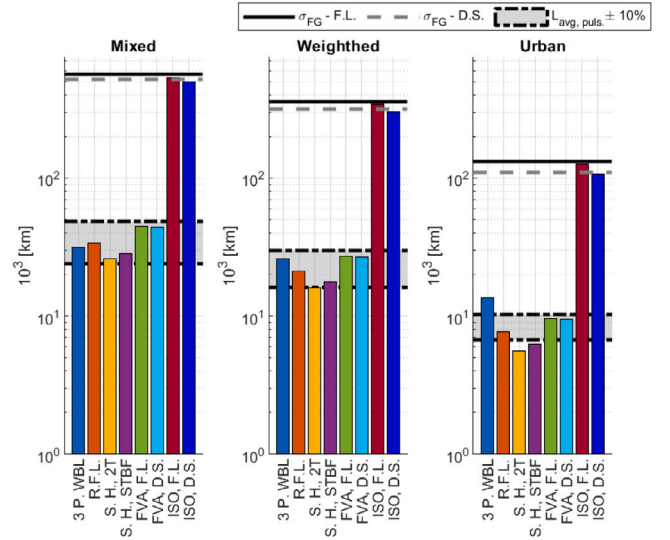


Fig. 15. Average estimated lifetimes.

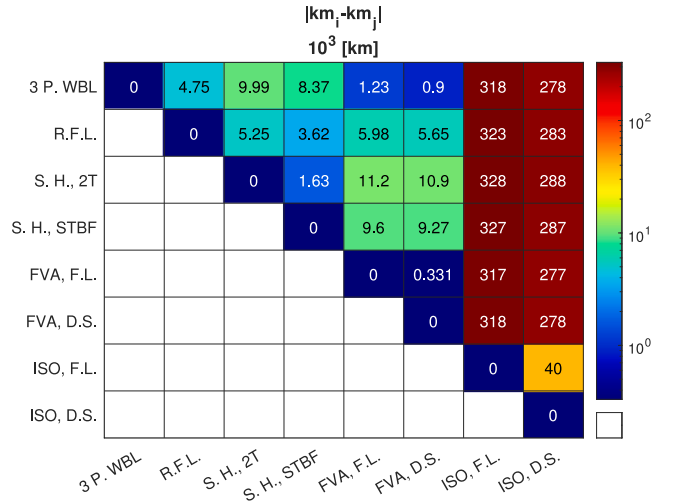


Fig. 16. Comparison between the different estimated lifetimes for the “Weighted” spectrum.

In order to reduce the effect of the different strength values between the campaigns, as well as with those of the gears reported in [1,41,42], for each experimental dataset, the spectra have been scaled in the  $\sigma$  direction according to the ratio of the experimental endurance limit according to FVA 304 [17] and the equivalent fatigue limit of the gears defined in the original sources [1,41,42] and calculated according to ISO 6336-3 [6]; an example of such scaling procedure is depicted in Fig. 7. The validity of this operation can be understood by looking at Figs. 8, 9 and 12–15. On the one hand, as shown in Figs. 8, 12 and 14 there is no substantial difference between the results for “ISO, F.L.” and “ISO, D.S.” curves for the different experimental campaign. The small difference between them is related to the different slopes associated with the limited life region. On the other hand, as reported in Fig. 9, Fig. 13, and Fig. 15, the same applies between the averaged results of “ISO, F.L.” and “ISO, D.S.” and the one according to ISO 6336-3 [6].

Thanks to the analysis of the pulsator data elaboration and their use within spectra-assessment (i.e. without considering the “ISO, F.L.” and “ISO, D.S.” curves), it is possible to notice that, depending on the adopted statistical model, the assessment provides a different outcome, as shown in Figs. 8, 12 and 14. However, it is not possible to identify a more conservative method that could provide a lower safety factor (or a lower estimated lifetime). Indeed, there is no fixed relative position of the results.

Regarding the same spectrum but different materials, it is possible to consider some examples, as “3 P. WBL” and “R.F.L.” for the materials A and E which, for Case 1 (i.e. Fig. 8), have a different relative position. For A material, the “3 P. WBL” safety factor is lower than the “R.F.L.” one, while for the E case, the result is the opposite. A similar example is shown for “S.H., STBF/2T” with respect to “FVA, F.L./D.S.” for material C and A safety factors in respect to the “Weighted” spectrum (i.e. Fig. 12). Also, the estimated lifetime (i.e. Fig. 14) shows this not-uniform relative positioning (see “R.F.L.” vs “S.H., STBF/2T” for material A and C).

In any case, the overall results are relatively similar to each other. Indeed, they all are within an interval equal to  $\pm 5\%$  the average safety factors (as reported in Figs. 9 and 13) while, as shown in Fig. 15, the estimated average lifetimes mostly are within a  $\pm 10\%$  interval (on a logarithmic scale). All the above suggests that, considering few selected spectra and few materials, the selection of a particular curve, with a more (or less) strict statistical theory behind itself, it seems to have not a particular effect for an assessment based on liner damage accumulation law.

The actual discrepancy is not between the different models interpreting the pulsator evidence but between the pulsator evidence per se and the curves obtained according to the standards (see Figs. 9, 10, 13, 15 and 16). Specifically, pulsator tests provide a more conservative outcome (i.e. lower safety factors or lower estimated lifetimes) than ISO 6336-3 [6], which, fittingly, considers the case of gears rotating under load. This difference is attributable to the different position of the fatigue knee that, for all the pulsator tests considered here, is located below  $5 \times 10^5$  cycles. According to the authors' experience and as shown in Fig. 7 and by the other cases reported in [43], such location does not vary significantly after pulsator data elaboration. However, ISO 6336-3 [6] (as well as ANSI AGMA 2001-D04 [2]), which – fittingly – consider only the case of meshing gears, poses the fatigue knee at  $3 \times 10^6$  cycles. Fig. 5 shows an example of this difference for the exemplary campaign B.

A physical effect of this difference comes from the interpretation of Fig. 14 results and from the differences highlighted by Fig. 16. Assuming that the load spectrum “Weighted” is representative of the actual working of a truck, its lifetime estimated on the basis of pulsator data is, as average, 22 thousand kilometers, far below what can be considered a typical lifetime. On the other hand, the curves based on ISO 6336-3 [6] report an estimated lifetime of around 250–350 thousand kilometers, values that are still relatively low but, surely, closer to what can be expected as a truck lifetime.

However, all the experimental evidence from pulsator tests collected in Table 2 suggests that pulsator tests show a fatigue knee well below  $1 \times 10^6$  load cycles. Among the 269 analyzed experimental campaigns, 18 show a fatigue knee at a number of cycles higher or equal to 1 million of cycles, but 248 at a number of cycles lower or equal to  $5 \times 10^5$ . Nevertheless, all the few publications reporting meshing gear tooth root fatigue tests with a running gear pair present a fatigue knee above  $1 \times 10^6$  [17,70,79]. Specifically, [17] reports one collective evaluation of 6 campaigns whose fatigue knee is in average at approx. 1 million cycles. [70] includes 5 campaigns whose fatigue knee is located between 2 and 4 million cycles and in [79] 3 campaigns whose fatigue knee is about 10 million cycles are presented. Values comparable with the 3 million suggested by the standards [2,6].

All the above suggests that there are intrinsic differences between pulsator and the meshing gear scenario that have not been considered so far. Hence, when the limited life region and its transition to the long life one is considered, the application of Rettig's factor [37] or any of the other methodologies developed so far is not enough, as other effects might be present but not considered.

One theory that may shed more light on this aspect is that the load sharing between teeth is affected by the crack propagation in the tooth root. Indeed, the presence of a crack modifies the tooth stiffness (e.g. [80–83]). Therefore, the load sharing between adjacent teeth pairs is altered, resulting in a reduction of the stresses acting on the damaged tooth root, lowering the crack propagation speed [84]. Clearly, this aspect is not present within pulsator tests, as the applied load acts always on the same teeth, without any load sharing.

Hence, a less conservative usage of pulsator test data within the load spectra would be the usage of pulsator tests only as a means to estimate the gear fatigue limit (i.e. ISO  $\sigma_{Flim}$  [6] or ANSI AGMA  $s_{at}$  [2]). Then, the SN curve is derived following what is predicted by the standards [2,6]. Indeed, their fatigue knee (at 3 million cycles) is closer to what is suggested by the running gears tests.

The rationales for such choice are, on the one hand, the calculated lifetimes for the Case 2 load spectra for the “ISO F.L./D.S.” curves, which are closer to a generic expected lifetime of a truck (see Figs. 14 and 15). On the other hand, a more complex rationale can be drawn by introducing Murakami's idea of non-propagating cracks, which suggests that the limited life and the fatigue limit (i.e. the long life) are related to two different aspects (e.g. [85–88]). Murakami's idea implies that, in the region ahead of the fatigue knee (i.e. the limited life region) the cracks nucleate and propagate until component/specimen failure, while, in the region after the fatigue knee, cracks can nucleate but their propagation can stop. Recent studies have shown that Murakami's idea holds also for tooth root fatigue failure [89–92]. The application of this idea in our case of interest suggests that the effect of different crack propagation speeds, related to the crack effect on the load sharing, only applies to the limited life region, not to the long life one (as here, the crack nucleate but does not propagate). Hence, pulsator tests can still be used to estimate the teeth fatigue limit, from which the fatigue limit of the gear can be derived.

In any case, it must be pointed out that further experimental campaigns on tooth root bending fatigue, performed both via pulsator test and via running gears on the same gear specimen, considering the same region of the SN curve, are recommended in order to analyze in detail their differences; aiming to extend the known pulsator test evaluation methods. Hence, reducing the discrepancies between this experimental method and the actual application scenario, where gears rotate under loads.

Regardless, pulsator tests remain an established and suitable methodology to determine strength values (long life) in a reliable and cost-efficient way, and can be adopted within standardized load-carrying capacity calculation methods (e.g. ISO 6336-3 [1]). Even though, as seen above, pulsator test results obtained in the limited life range cannot be transferred directly to meshing gear conditions without proper and extended experience. In any case, this kind of methodology

**Table 2**

Fatigue knee position for different pulsator test campaigns on case-hardened gears. All the campaigns differ in terms of gear materials, heat and surface treatments. Data refer to the 50% percentile. The value reported for [17] is an average data coming from a collective evaluation. Data from [32] are only those at ambient temperature.

| Ref. | Year | Total number of points | Number of campaigns | Fatigue knee location                                                          |
|------|------|------------------------|---------------------|--------------------------------------------------------------------------------|
| [64] | 1966 | 177                    | 16                  | 14 between $10^5$ and $5 * 10^5$ , 2 between $5 * 10^5$ and $10^6$             |
| [65] | 1986 | 171                    | 2                   | 1 at $\approx 5 * 10^5$ , 1 at $\approx 10^6$                                  |
| [66] | 1989 | 265                    | 18                  | $\approx 10^5$ (all)                                                           |
| [67] | 1993 | 392                    | 9                   | 2 below $5 * 10^5$ , 7 above $10^6$                                            |
| [68] | 1994 | 215                    | 4                   | 1 at $\approx 10^6$ , 1 at $\approx 5 * 10^5$ , 2 at $\approx 2 * 10^5$        |
| [69] | 1996 | 104                    | 1                   | $\approx 3 * 10^6$                                                             |
| [17] | 1999 | 2219                   | 93                  | $\approx 5 * 10^5$                                                             |
| [70] | 2004 | $\approx 350$          | 16                  | 4 above $1 * 10^6$ , 10 between $1 * 10^5$ and $5 * 10^5$ , 2 below $1 * 10^5$ |
| [71] | 2004 | 28                     | 2                   | $\approx 7 * 10^5$ (all)                                                       |
| [72] | 2007 | 56                     | 4                   | $< 2 * 10^5$ (all)                                                             |
| [73] | 2010 | 56                     | 4                   | 1 at $9 * 10^4$ , 3 at $2 * 10^5$                                              |
| [21] | 2013 | 4                      | 1                   | $6 * 10^5$                                                                     |
| [74] | 2015 | No info                | 4                   | 4 below $5 * 10^5$                                                             |
| [75] | 2016 | No info                | 2                   | 1 at $9 * 10^4$ , 1 at $3 * 10^5$                                              |
| [13] | 2017 | 32                     | 1                   | $\approx 8 * 10^4$                                                             |
| [22] | 2017 | 4                      | 2                   | 2 at $2 * 10^5$                                                                |
| [25] | 2018 | 5                      | 5                   | 1 at $3 * 10^6$ , 1 $\approx 1 * 10^6$ , 3 below $3 * 10^5$                    |
| [76] | 2019 | 28                     | 1                   | $< 10^5$                                                                       |
| [43] | 2022 | 1643                   | 76                  | $\approx 5 * 10^5$ (all)                                                       |
| [32] | 2022 | $\approx 200$          | 6                   | 1 above $3 * 10^6$ , 1 $\approx 6 * 10^5$ , 4 below $5 * 10^5$                 |
| [77] | 2022 | 21                     | 1                   | $3 * 10^5$                                                                     |
| [78] | 2023 | 40                     | 1                   | $\approx 1 * 10^6$                                                             |

can still be adopted for comparative studies examining the limited life range of a gear. It is possible to consider examples like research investigating the effects of design variants (e.g. geometries, materials and treatments) on overload sensitivity.

## 9. Conclusion

Within the wider context of the assessment of a gear pair with respect to tooth root fatigue fracture, this article discusses the use of data obtained from pulsator tests (i.e. experimental tests performed on the teeth) when load spectra are considered in the design phase. Here, the starting point is a series of experimental tests, several different load spectra and gear with known geometries and materials.

The tests have been analyzed using several statistical methods and then applying corrective literature factors, obtaining from the tests performed on teeth different experimental curves referred to the gear wheel, in meshing gear condition, and at a 99% reliability level. The adopted methodologies represent, so far, the state of the art. The assessments have been conducted within the context of ISO 6336-6 [1], that is adopting a linear damage accumulation in accordance with the Palmgren-Miner rule, thus calculating a safety factor with respect to the applied spectrum as well as the estimated lifetime. For the purpose of reducing the differences among the various tests, the tests themselves and the starting material, the spectra have been scaled. Several spectra have been taken into account, one referring to a gearbox of a lifting machinery and three measured on a reduction stage of a truck transfer box.

Focusing on the analysis of the curves obtained from the pulsator tests, it has been observed that there is not a substantial impact of the methodologies associated with the test. In fact, there is not a particular result in terms of the relative positioning of the curves and, as a matter of fact, the assessment, both in terms of safety coefficient and estimated lifetime, provides results that are comparable and not significantly different.

The real discrepancy is not between the values provided by the elaborated pulsator tests *per se* but between those curves derived from them and the curves obtained following ISO standards. The rationale is related to the dissimilar location of the fatigue knee which turns out to be positioned below 500,000 cycles for the pulsator tests, while the standards (i.e. [2,6]) provide it at 3 million cycles. This discrepancy is not related to the pulsator tests used here, but it is a general behavior of

such testing methodology, which provides curves whose knee is quite distant from what the experimental evidence on meshing gear suggests being representative of tooth root fatigue fracture.

Therefore, the designer who wants to use pulsator tests as the base of a spectrum test must be aware that, given the above differences, such test methodology does indeed provide specific test data but that such data are conservative with respect to the actual scenario, where gears are rotating under load.

## Nomenclature

| Symbol                  | Meaning                                                        | Source  |
|-------------------------|----------------------------------------------------------------|---------|
| $B$                     | Threshold parameter for lifetime                               | [58]    |
| $C$                     | Threshold parameter for stress level                           | [58]    |
| $f_{P \rightarrow MG}$  | Rettig's corrective factor                                     | [17,37] |
| $k_1$                   | Limited life region slope                                      | [56]    |
| $k_2$                   | Long life region slope                                         | [56]    |
| $N_e$                   | S.H. model coordinate (together with $\sigma_e$ )              | [56]    |
| $N_i$ or $N_{\sigma_i}$ | Maximum number of cycles associated to the $i$ th load bin     | [1]     |
| $n_i$                   | Number of cycles associated to the $i$ th load bin             | [1]     |
| $N_r$                   | Number of repetitions                                          | [1]     |
| $S$                     | Safety factor for load spectra assessment                      | [1]     |
| $s$                     | R.F.L. std. dev. of the conditional distribution               | [59]    |
| $s_{at}$                | Allowable bending stress number                                | [2]     |
| $s_{1, \log(N)}$        | S.H. std. dev. in the $\log(n)$ direction, limited life region | [43,56] |
| $s_{2, \log(N)}$        | S.H. std. dev. in the $\log(n)$ direction, long life region    | [43,56] |
| $s_\gamma$              | R.F.L. std. dev. of the fatigue limit distribution             | [59]    |
| $U$                     | Damage                                                         | [1]     |
| $V$                     | Logarithm of the fatigue limit $\gamma$ in the R.F.L. model    | [59]    |
| $W$                     | Logarithm of the fatigue life $Y$ in the R.F.L. model          | [59]    |
| $X$                     | Distribution estimated from pulsator test evidence (SoE)       | [93]    |

|                              |                                                                            |      |
|------------------------------|----------------------------------------------------------------------------|------|
| $x$                          | Logarithm of the stress level $\sigma$ in the R.F.L. model                 | [59] |
| $Y$                          | R.F.L. fatigue life                                                        | [59] |
| $Y_{NT}$                     | Life factor for tooth root stress                                          | [6]  |
| $Y_{RelT}$                   | Relative surface factor                                                    | [6]  |
| $Y_{ST}$                     | Stress correction factor                                                   | [6]  |
| $Y_X$                        | Size factor relevant to tooth root strength                                | [6]  |
| $Y_{\delta relT}$            | Relative notch sensitivity factor                                          | [6]  |
| $Z$                          | Number of extractions of $X$ (SoE)                                         | [93] |
| $z$                          | Number of teeth                                                            | –    |
| $\beta$                      | Shape parameter of the Weibull model                                       | [58] |
| $\beta_0$                    | R.F.L. curve parameter                                                     | [59] |
| $\beta_1$                    | R.F.L. curve parameter                                                     | [59] |
| $\gamma$                     | R.F.L. curve fatigue limit parameter                                       | [59] |
| $\delta$                     | Scale parameter of the Weibull model                                       | [58] |
| $\epsilon$                   | R.F.L. curve error term                                                    | [59] |
| $\theta$                     | Vector of R.F.L. model parameters                                          | [59] |
| $\lambda$                    | Threshold parameter of the 3 parameters Weibull model                      | [58] |
| $\mu$                        | Mean value of a generic distribution                                       | –    |
| $\mu_\gamma$                 | R.F.L. mean value of the fatigue limit distribution                        | [59] |
| $\sigma$                     | Stress value                                                               | –    |
| $\sigma_e$                   | S.H. model coordinate (together with $N_e$ )                               | [56] |
| $\sigma_{(F0\infty,1\%,MG)}$ | Gear endurance limit for 1% failure probability in meshing gear conditions | [17] |
| $\sigma_{(F0\infty,50\%,P)}$ | Experimental (pulsator) endurance limit for 50% failure probability        | [17] |
| $\sigma_{Flim}$              | Allowable stress number                                                    | [6]  |
| $\sigma_i$                   | Load level of the $i$ th load bin                                          | [1]  |
| $\Phi_{V W}$                 | CDF of the conditional distribution of $V   W$ in the RFL model            | [59] |
| $\phi_V$                     | pdf of the fatigue limit in the R.F.L. model                               | [59] |

## CRedit authorship contribution statement

**Luca Bonaiti:** Conceptualization, Data curation, Methodology, Software, Writing – original draft, Writing – review & editing. **Marco Monti:** Conceptualization, Data curation, Methodology, Software, Writing – original draft, Writing – review & editing. **Michael Geitner:** Conceptualization, Data curation, Methodology, Writing – original draft, Writing – review & editing. **Thomas Tobie:** Data curation, Supervision, Writing – original draft, Writing – review & editing. **Carlo Gorla:** Methodology, Supervision, Writing – original draft, Writing – review & editing. **Karsten Stahl:** Data curation, Supervision, Writing – original draft, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Appendix. Models description

This appendix reports a short description of the several statistical models included in this article. Furthermore, also the statistical passage from the pulsator tests to the gear is depicted as well.

The first model adopted here is the FVA 304 [17], which elaborated the pulsator test evidence on the basis of the typical scatter observed

in gear testing. In this model, the limited life region and the long life region are estimated differently and then joined. The limited life region of the gear is defined by a line fitting of lifetime observed during the tests, corrected at the proper gear reliability level. On the other hand, the long life region is described by an horizontal line. Here,  $\sigma_{(F0\infty,50\%,P)}$ , which is the experimental endurance limit estimated according to [94] or [95], is corrected by means of coefficients in order to define  $\sigma_{(F0\infty,1\%,MG)}$ , which is the gear endurance limit for 1% failure probability.

The second adopted model is the one proposed by Spindel and Haibach [56] (here designated as S.H.). Accordingly the random variable (i.e. the number of cycles) can be described by means of a lognormal distribution, whose means depends on the applied stress level.

$$\log(N) \sim \mathcal{N}_{(\mu_{(\log(\sigma))}, s)} \quad (\text{A.1})$$

The mean is defined by a double slope curve (i.e. (A.2)) and a constant standard deviation in the  $\log(\sigma)$  direction which results in two different standard deviations in the  $\log(N)$  direction  $s_{1,\log(N)}$  and  $s_{2,\log(N)}$ .

$$\begin{aligned} \mu_{(\log(\sigma))} = \log(N) = \log(N_e) + \frac{1}{2} (k_1 + k_2) (\log(\sigma) - \log(\sigma_e)) \\ + \frac{1}{2} (k_1 - k_2) |\log(\sigma) - \log(\sigma_e)| \end{aligned} \quad (\text{A.2})$$

where  $N_e$  and  $\sigma_e$  are the coordinates of the knee,  $k_1$  is the slope associated with the limited life region and  $k_2$  is the one associated with the long life region. As this model allows the usage of randomly censored data, the curve can be referred to both the experimental data (i.e. the “STBF” curve) or to the teeth (i.e. the “2T” curve).

The third model is the Random Fatigue Limit (R.F.L.) one and has been proposed by Pascual and Meeker [59]. It makes use of a curve with a smooth transition between the finite life region and the fatigue limit, without a knee point. In this model, the fatigue life  $Y$  was described as

$$\log(Y) = \beta_0 + \beta_1 \log(\sigma - \gamma) + \epsilon, \quad \sigma > \gamma, \quad (\text{A.3})$$

where  $\beta_0$  and  $\beta_1$  are the fatigue curve coefficients,  $\gamma$  is the fatigue limit and  $\epsilon$  is the error term. This correlation holds only for stress levels higher than the fatigue limit, which can be modeled either with a smallest extreme values (sev) or a normal distribution with mean value  $\mu_\gamma$  and standard deviation  $s_\gamma$ .

Considering  $V = \log(\gamma)$ ,  $x = \log(\sigma)$ ,  $W = \log(Y)$ , it can be assumed that, conditioned on a fixed value of  $V < x$ , the probability density function (pdf) of  $V | W$  is

$$f_{V|W}(w; \beta_0, \beta_1, s, x, v) = \frac{1}{s} \phi_{V|W} \left( \frac{w - [\beta_0 + \beta_1 \log(e^x - e^v)]}{s} \right), \quad (\text{A.4})$$

with location parameter  $\beta_0 + \beta_1 \log(e^x - e^v)$  and scale parameter  $s$ .  $\phi_{V|W}(\cdot)$  can be either the normal or the standardized sev pdf. In this paper, the normal distribution was used both for  $\phi_V$  and  $\phi_{V|W}$ .

Hence, the Cumulative Distribution Function (CDF)  $F$  can be defined as:

$$F(w; x, \theta) = \int_{-\infty}^x \frac{1}{s_\gamma} \Phi_{V|W} \left( \frac{w - \mu(x, v, \theta)}{s} \right) \phi_V \left( \frac{v - \mu_\gamma}{s_\gamma} \right) dv, \quad (\text{A.5})$$

where  $\theta = (\beta_0, \beta_1, s, \mu_\gamma, s_\gamma)$ ,  $\mu(x, v, \theta) = \beta_0 + \beta_1 \log(e^x - e^v)$  and  $\Phi_{V|W}(\cdot)$  is the CDF of  $V | W$ .

Therefore, this model adopts two different distributions which allow to describe separately the life and the fatigue limit. The CDF of this model is therefore constituted by two different contributions: the pdf related to the fatigue limit and the conditional CDF of the life associated to each stress level higher than the fatigue limit.

The last model is the one proposed by Castillo and Fernández-Canteli, who performed a Buckingham theorem analysis in order to determine the relationship between number of cycles and applied load. Among the models proposed in [58], the 3 Parameters Weibull model

has been selected; here this model is called “3P. WBL”. Accordingly, the SN field can be described as:

$$F_{(N,\sigma)} = 1 - \exp \left[ - \left( \frac{(\log(N) - B)(\log(\sigma) - C) - \lambda}{\delta} \right)^\beta \right] \quad (\text{A.6})$$

where  $B$  is the threshold parameter for lifetime.  $C$  is the threshold parameter for stress level (and corresponds to the endurance limit),  $\lambda$  is the threshold parameter of the 3 parameters Weibull model,  $\delta$  and  $\beta$  are the scale parameter and the shape parameter of the Weibull model.

The “R.F.L.” and “3P. WBL” have been used only to describe the experimental curve, that is composed by teeth pairs.

For the “S.H.”, “R.F.L.” and “3P. WBL” models, the passage from the pulsator test data is done under the hypothesis that a gear failure due to tooth root fatigue failure is ruled by the fracture of its weakest teeth (e.g. [15,31,36,37,52]). This passage is done by means of SoE. Indeed, through a mathematical passage, the CDF of the minimum value observed on  $Z$  extractions of  $X$  (whose distribution is known) can be calculated. In our case CDF of the minimum value represents the gear and  $N$  is the  $z$  number of teeth (or the  $z/2$  pairs of teeth, if the curve refers to experimental data) and  $X$  is the distribution estimated from the pulsator test evidence (i.e. Eqs. (A.1), (A.5) and (A.6)).

$$\begin{cases} F_{\text{gear}} = 1 - (1 - F_{\text{teeth}})^z \\ F_{\text{gear}} = 1 - (1 - F_{\text{pulsator}})^{\frac{z}{2}} \end{cases} \quad (\text{A.7})$$

Here, this mathematical passage is called SoE following the nomenclature proposed in [93]. This kind of equation can also be used in order to describe the failure probability of system in series [96].

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